Prabin Karki

PhD Candidate in Physics (Biophysics), Department of Physics, University of Wisconsin–Milwaukee, Milwaukee, WI 53211

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Summary

Structural biologist with three and a half years of experience determining macromolecular structures by cryo-EM and X-ray crystallography, including time-resolved methods. First author of three papers and preprints on bacteriophytochrome signaling. Strong background in biophysics, biochemistry and molecular biology. Seeking to apply structural biology to protein-focused research and drug discovery.

Skills

- **Cryo-EM:** single-particle and time-resolved cryo-EM, Spotiton grid preparation; cryoSPARC, RELION, crYOLO, cryoDRGN
- **Crystallography:** serial and time-resolved X-ray crystallography at synchrotrons and XFELs; CrystFEL, Phaser, REFMAC5, CCP4
- **Model building and refinement:** ChimeraX, ISOLDE, Coot, Phenix; interpreting protein dynamics from structure
- **Wet lab:** cell culture, protein overexpression and purification, chromatography, SDS-PAGE, characterization assays
- **Computing:** Python and bash scripting, custom software, data-processing pipelines on high-performance computing clusters
- **Instrumentation:** design and development of custom experimental setups
- **Communication:** scientific writing, explaining complex topics to general audiences

Education

- **Ph.D. in Physics** (Biophysics focus), University of Wisconsin–Milwaukee — GPA 3.7 — 2022–present
- **M.Sc. in Physics**, Tribhuvan University, Nepal — GPA 3.55 — 2021
- **B.Sc. in Physics**, Tribhuvan University, Nepal — 2014

Research experience

Graduate Research Assistant, Schmidt Lab, Department of Physics, UWM — 2023–present

- Lead the project on the bacteriophytochrome–response regulator complex, from protein production to cryo-EM structure.
- Lead author on multiple cryo-EM studies of phytochromes, including single-millisecond time-resolved cryo-EM with the Spotiton technique.
- Collected and analyzed data for collaborative projects on phytochrome and rhodopsin dynamics, including time-resolved crystallography at synchrotrons and XFELs.
- Installed, migrated and maintained the lab's data-processing pipeline on a high-performance computing cluster.

Teaching and mentoring

Data-processing workshop organizer, Department of Physics, UWM — 2025

- Proposed, planned and ran a full-day workshop training biophysics postdocs and graduate students in crystallographic and cryo-EM data processing.
- Lectured and led the hands-on sessions on bash scripting, cluster computing and data analysis.

Teaching Assistant, Department of Physics, UWM — 2022–2024

- Physics I, calculus-based (Spring 2023, Fall 2023): led two weekly discussion sessions, graded quizzes and homework, wrote practice problems and mock exams.
- General Physics I, non-calculus (Spring 2024): led two weekly discussion sessions, graded quizzes and homework, wrote practice problems and mock exams.

Fellowships and awards

- Graduate Student Excellence Fellowship, UWM — 2026–2027
- James Martin Memorial Scholarship, UWM — 2026–2027
- Physics Research Excellence Award, Department of Physics, UWM — 2024–2025
- Chancellor's Graduate Student Award, Department of Physics, UWM — 2022–2026
- Master's Thesis Grant, Government of Nepal — 2017–2018

- Most Deserving Student Scholarship, Tribhuvan University — 2012–2013

Peer-reviewed publications

1. **Karki P.**, Budell W. C., Kannachel R., Menendez D., Hernandez C., Mendez J. H., Kanou S., Singh M., Slavov C., Schwander P., Malla T. N., Stojković E. A., Schmidt M. A bacteriophytochrome Pr/Pfr heterodimer studied through single-particle time-resolved cryo-electron microscopy. *Communications Chemistry* (2026).
2. **Karki P.**, Menendez D., Budell W., Dangi S., Hernandez C., Mendez J., Muniyappan S., Basu S., Schwander P., Malla T. N., Stojković E. A., Schmidt M. Structures of myxobacterial phytochrome revealed by cryo-EM using the Spotiton technique and with X-ray crystallography. *Structural Dynamics* 12, 034701 (2025).
3. Malla T. N., Muniyappan S., Menendez D., Ogukwe F., Dale A. N., Clayton J. D., Weatherall D. D., **Karki P.**, Dangi S., Mandella V., Pacheco A. A., Stojković E. A., Rose S. L., Orlans J., Basu S., de Sanctis D., Schmidt M. Exploiting fourth-generation synchrotron radiation for enzyme and photoreceptor characterization. *IUCrJ* 12, 36–48 (2025).

Preprints and manuscripts under review

1. **Karki P.**, Hernandez C., Mendez J. H., Budell W. C., Menendez D., Kanou S., Razavi S., Espoukeh P., Abuhashish R., Bizhga D., Basu S., Malla T. N., Schwander P., Stojković E. A., Schmidt M. The signaling pathway of a bacterial two-component system determined by cryo-EM. *bioRxiv* (2026). Under peer review.
2. Meszaros P., Grunewald L., Prodhan S., Monrroy L., Zhou Y., Scattolini G., Bodizs S., Chowdhury S., Larkiala T., Gautier C., Voloshyn I., Wahlgren W. Y., **Karki P.**, Malla T. N., Luo F., Kang J., Owada S., Nango E., Tono K., Beale E. V., Bacellar C., Dworkowski F., Cirelli C., Johnson P., Beale J. H., Appleby M., Takala H., Schmidt M., Maj M., Westenhoff S. Structural photoactivation of a bathy phytochrome resolved by time-resolved X-ray crystallography. Under review at *Science*.